Automatic Rain Detection and Protective Sheet With Artificial Light For Crops

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Abstract --Agriculture is among the leading food sources. However, this source of food has greatly relied on the prevailing weather. In some cases, sudden and heavy rainfalls during the rainy seasons lead to the formation of stagnant water around the farms. Through this, the farming yield may reduce. However, the effects of climate change have increased the occurrence of such weather . In this proposed project, an automatic rainfall detection and protection of the crops system is designed. Rain sensors are fixed on poles around the farmlands to continuously detect rainfall. As soon as rainfall is detected, the system automatically shades the farmlands using protection sheets to prevent buildup. In addition to this, the solar panel fixed in this helps store energy during the day. This energy is used to power the whole system and provides light to the farmlands during rainy days when the sun is not sufficient.
The system will be beneficial in preventing damage to crops by rain, enhancing production levels, and sustaining the use of renewable energy sources for farming.

Keywords— Rain sensor and protection sheet, , Internet of Things, solar power, Artificial light , Servo Motor, Arduino uno

I. INTRODUCTION

Agriculture is the foundation for food security and economic development, especially in developing countries. But farmers often struggle under very tough conditions due to uncertain weather. Out of which, heavy rain is one of the key factors resulted in crops damage, waterlogging and unproductivity of agriculture. Soil saturation from excess rainfall reduces oxygen available for plant roots, and leads to nutrient leaching. Conventional farming techniques do not offer a quick solution to abrupt weather changes, and available alarm systems are often left unresponsive in real-time. Farmers therefore face massive economic losses.

However, with developments in Internet of Things (IoT) technologies, it is feasible to create intelligent systems that can monitor environmental conditions and respond accordingly. Whereas IoT provides access to the data immediately and assists in decision-making via interactions of sensors and devices . In this paper, rain sensors and motorized protective sheets are combined to propose a smart automated system for crops protection against heavy rain. The address in a proposed model is also integrated with solar panels for renewable energy generation and artificial lighting when there are several cloudy or rainy days unlike conventional systems. This enables crops to grow continuously, even in low sunlight environments.

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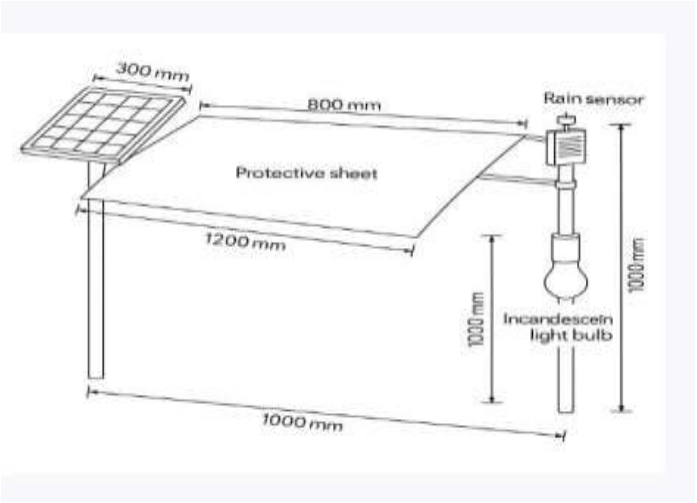


Fig. 1. Crop protection system and solar-powered mechanism.

The system makes use of the rain sensor, which helps in monitoring the level of rainfall at all times. It senses the water droplets falling and produces a signal, which is used by the microcontroller unit that controls the operations of the whole system. Upon sensing the rain, the microcontroller turns on the motor-driven system, which releases the protection sheet to cover the crops from excessive water. The system employs the solar energy power system, which consists of the solar panel, battery, and charge controller. The energy obtained from the solar panel is used to drive the system's control systems and any other component. Apart from using the solar energy system, there is also another system of lighting made up of LED. This will help illuminate the crops during rainy days with little sun availability.

With the use of sensors, automation, and renewable energy sources, the system becomes energy-efficient and sustainable for use in agriculture. Microcontroller is used as the central processor because of its easy-to-program and low-energy consumption nature and is realized through the use of the Arduino platform with C/C++ programming. A rain sensor module like the FC-37 or YL-83 enables accurate detection of rainfall and triggers timely actions in the system. After rainfall is detected, the control process actuates the motor system to spread the protective cover, whereas the light from the solar-powered lights is utilized when sunlight is limited.

II. RELATED WORK

The current developments in the field of smart agriculture have led to the creation of advanced solutions that aid farmers in safeguarding their produce from the adverse effects of the environment, such as rainfall and erratic weather patterns. The traditional practices of covering the fields manually, installing drainage facilities, and erecting temporary barriers are common practices but lack efficiency, consume a lot of time, and are labour-intensive.

The advent of IoT technology means that the use of sensors and embedded systems is employed in contemporary agriculture to monitor environmental conditions on-the-go. There are rain sensors, moisture sensors in the soil, and weather monitoring system that allow farmers to track any changes in the environment and receive notifications about these. The above systems enhance farmers' ability to observe the environment, but the majority of such systems do not provide any action towards the protection of crops.

There are several automatic systems that protect crops from being destroyed by rain through motorization of a protective device that covers crops. These systems may offer protection from rainfall to some extent; however, they still use an external power source. Most systems ignore the problems of excessive cloudiness that might be detrimental to crop development because of lack of sun exposure.

Current studies also focus on the implementation of microcontroller systems including Arduino and embedded systems to achieve cheap and convenient automation in farming. The systems support real-time analysis of data collected by sensors and manipulation with actuators such as motors and relays. Nevertheless, their functionality is limited due to the lack of integration with solar energy sources and artificial light.These limitations are overcome using the system that combines the elements of rain detection, protection of crops, utilization of solar energy, and artificial light in one efficient structure. It allows not only to protect the products from the excessive rains but also to provide the process of cultivation even when there is no natural light available.

microcontrollers, and any remote device or Internet service could control those parameters. Power supply and generation were frequently major issues. This initiative used technology-based solutions for crop safety as well as solar power generation and retention ponds.

Bansod et al. created a model regarding crop protection from heavy rainfall [10]. The authors incorporated the soil moisture sensor, which was activated when water between two electrodes completed the circuit and provided a signal to the controller. The controller then instructed the motor driver to cover the chosen field area with cloth or plastic. The NodeMCU module provided the automated system with total control. The connection of the module to a browser gave the farmer a live reading of the temperature and humidity as well as an ON/OFF switch. The recommended architecture addressed how to automatically protect the field from rain using an Arduino Uno as the controller. They could prevent the devastation of tens of thousands of hectares of agriculture with this planned framework. If the amount of water in the soil is insufficient, extra rainwater can be retained and used for residential purposes as well as crops.

III. METHODOLOGY

A. Rain Detection Unit:

The rain sensor module detects rainfall through its installation within the farm. The sensor recognises any water droplets that form on its surface and keeps track of the surrounding environment continuously. This becomes an invasive monitoring process that detects rainfall immediately. The readings from the sensor are relayed directly to the processing unit.

B. Processing Unit

This is done using the microcontroller that is built into the device, the Arduino UNO. This controller constantly monitors the input from sensors and performs the control operations depending on certain set criteria. On detection of rainfall, it initiates certain processes that protect the crops.

C. Control and Actuation Unit

The control signals produced by the microcontroller help to actuate the actuator parts. In order to move the protective sheet, a servo motor is required. The motor moves the sheet during the rainy season in order to avoid water deposition on the crops. Once the rain ceases, the motor again pulls the sheet in order to bring back normally.

D. Power and Energy Unit

The system runs on electricity generated by a combination of a solar panel with a battery. This system works by converting energy from the sun into electric current using the solar panel, and this energy is stored in the battery to ensure that the system runs smoothly.

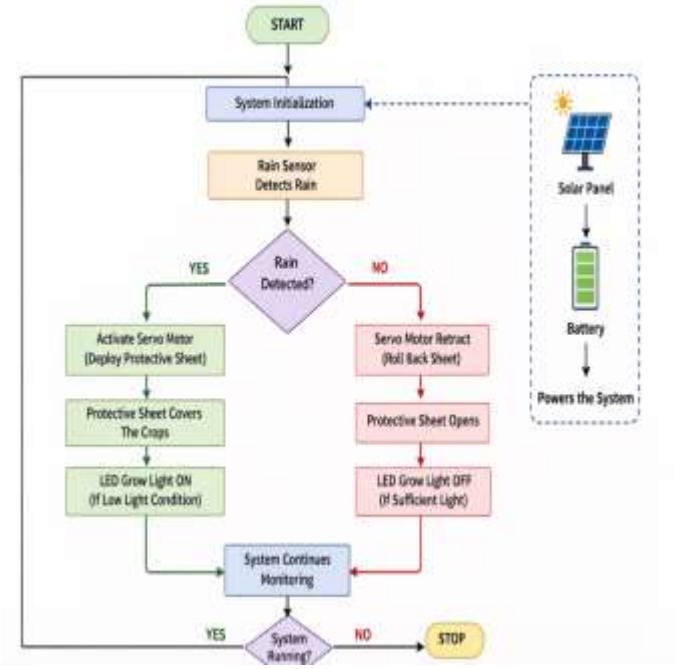


Fig. 2. Work model of protection system and solar power.

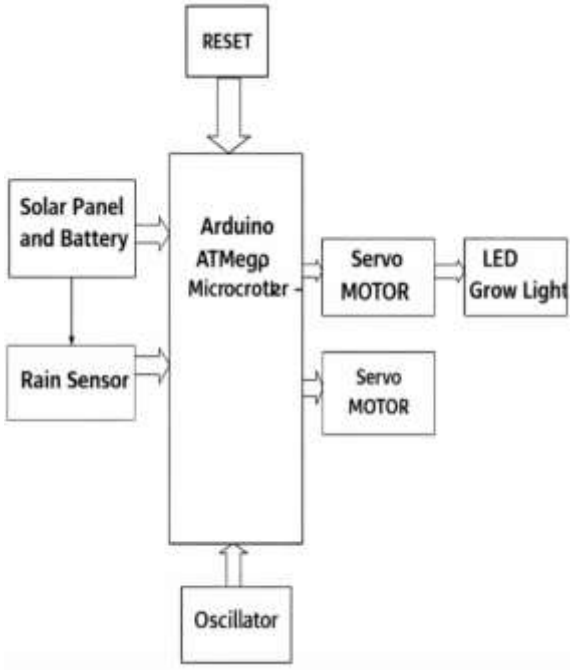


Fig. 3. Block Diagram of automatic rain alert and protection system.

E. Lighting

An LED lighting system has been included to ensure adequate lighting in case of extended periods of cloudiness and rain. In cases where there is inadequate sunlight, the microcontroller switches on the LED lights.The system may also include alert mechanisms to notify farmers about rainfall conditions and system activity.

IV. RESULTS AND ANALYSIS

This system was tested in an artificially created farm environment so as to be able to experience varying levels of rainfall that can impact crop growth. The test was conducted to check the ability of the system to sense rain and trigger the protective action. Based on the findings of the tests, it has been established that the system is efficient enough to sense rain and initiate the protection system so as to avoid any build-up of water in the crops. The other feature of turning on the LED lights has also proven successful.

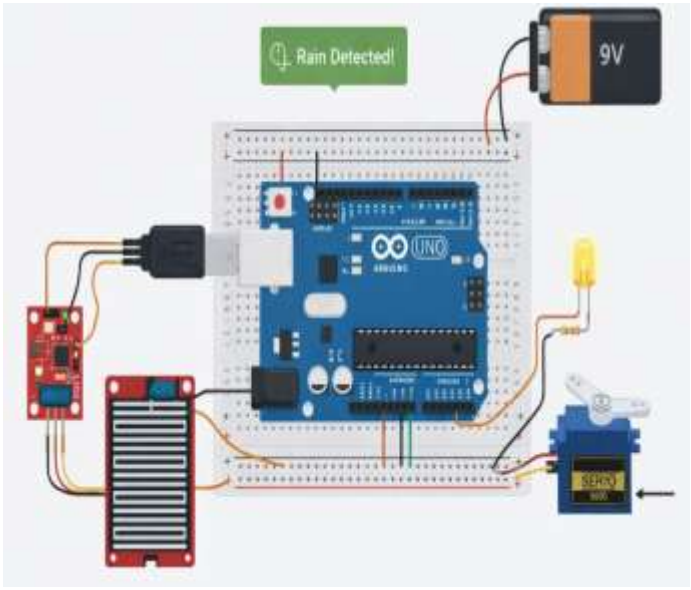


Fig. 4: simulation setup

The proposed system has been simulated to demonstrate its proper functioning for various environmental situations. Figure 4 depicts the simulation process, which involves the use of Arduino, rain sensor, servo motor, and LED lighting system. In this simulation, the input from rain sensor has been altered to demonstrate rainfall situation. During rainfall, the proposed system operated in such a manner that the servo motor started working, thereby covering the crops using the protective sheet. In addition to this, the LED indicator turned ON during rainfall situation to ensure artificial lighting when there is lack of light. On the other hand, during the absence of rainfall, the servo motor again returned to its initial position while LED indicator remained OFF.

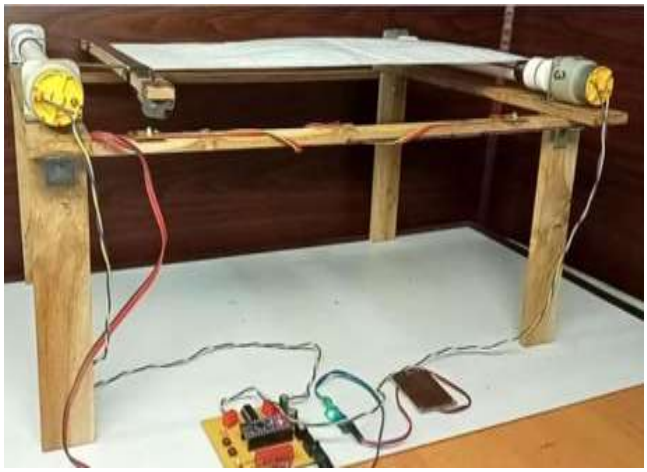


Fig. 5: simulation setup

Fig. 5 illustrates the hardware architecture of the proposed system. The hardware prototype is constructed through the use of a microcontroller, motor arrangement, and other components to illustrate the ability of the proposed system to protect crops from rainfall in real time. This design can be viewed as an idealized representation of a farm that incorporates the operation of the protective sheet controlled by the motors attached to the support frame. As shown in the experiments, the proposed system can recognize rainfall events and trigger the operation of the motor arrangement to cover the crops. In particular, the protective sheet works effectively in preventing the accumulation of water on crops. After the rainfall event, the sheet is retracted and everything returns to normal.

Parameter	Value / Observation
Rain Detection Method	Analog Sensor (A0)
Threshold Value	400
Response Time	2 – 5 seconds
Servo Motor Action	0° (Open), 90° (Close)
LED Operation	ON (Rain), OFF (No Rain)
Detection Accuracy	95%
System Operation	Automatic
Power Source	Solar Panel + Battery
System Reliability	92%

TABLE I: PERFORMANCE METRICS OF PROPOSED SYSTEM

V. CONCLUSION AND FUTURE WORK

Development of an IoT based automation control system which would help in overcoming the issues faced due to heavy rainfall and low light intensity in farming. The system uses a rain sensor to measure rain intensity and a motor-driven system is used to cover crops with a protective layer so as to save them from any kind of harm. A solar energy system has been incorporated into the system to ensure efficient working. An LED lighting system is employed in the system to support the growth of crops when light intensity is low or there are cloudy or rainy days.

In future, the system can be upgraded to incorporate all the functionalities of IoT such as wireless networking and data monitoring from the cloud along with the ability to control the system remotely using mobile applications. The use of weather forecasting techniques as well as intelligent decision-making systems will help further enhance the efficacy of the system. Furthermore, the system can also be deployed in agriculture on a larger scale with the aid of sensor technology and automation.

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